```
!unzip /content/mitdb.zip

Archive:  /content/mitdb.zip
   creating: mitdb/
  inflating: mitdb/209.atr
  inflating: mitdb/208.dat
  inflating: mitdb/109.hea
  inflating: mitdb/117.dat
  inflating: mitdb/105.xws
  inflating: mitdb/233.atr
  inflating: mitdb/220.xws
  inflating: mitdb/222.atr
  inflating: mitdb/201.xws
  inflating: mitdb/101.atr
  inflating: mitdb/234.dat
  inflating: mitdb/112.dat
  inflating: mitdb/215.atr
  inflating: mitdb/116.atr
  inflating: mitdb/107.dat
  inflating: mitdb/106.xws
  inflating: mitdb/115.dat
  inflating: mitdb/RECORDS
  inflating: mitdb/231.xws
  inflating: mitdb/217.dat
  inflating: mitdb/213.atr
  inflating: mitdb/233.xws
  inflating: mitdb/223.xws
  inflating: mitdb/117.hea
  inflating: mitdb/119.hea
  inflating: mitdb/113.atr
  inflating: mitdb/203.hea
  inflating: mitdb/205.atr
  inflating: mitdb/215.hea
  inflating: mitdb/203.at_
  inflating: mitdb/213.dat
  inflating: mitdb/232.atr
  inflating: mitdb/102-0.atr
  inflating: mitdb/208.hea
  inflating: mitdb/231.atr
  inflating: mitdb/233.hea
  inflating: mitdb/203.atr
  inflating: mitdb/212.hea
  inflating: mitdb/207.xws
  inflating: mitdb/102.dat
  inflating: mitdb/121.dat
  inflating: mitdb/116.dat
  inflating: mitdb/203.xws
  inflating: mitdb/108.atr
  inflating: mitdb/202.hea
  inflating: mitdb/111.hea
```

```
  inflating: mitdb/111.xws
  inflating: mitdb/202.atr
  inflating: mitdb/111.dat
  inflating: mitdb/118.hea
  inflating: mitdb/121.hea
  inflating: mitdb/220.hea
  inflating: mitdb/230.atr
  inflating: mitdb/222.dat
  inflating: mitdb/117.atr
 extracting: mitdb/ANNOTATORS
  inflating: mitdb/107.atr
  inflating: mitdb/112.atr
  inflating: mitdb/210.dat
  inflating: mitdb/217.atr
  inflating: mitdb/219.hea
  inflating: mitdb/108.at_
  inflating: mitdb/109.xws
  inflating: mitdb/207.hea
  inflating: mitdb/112.hea
  inflating: mitdb/203.at-
  inflating: mitdb/104.atr
  inflating: mitdb/103.xws
  inflating: mitdb/215.xws
  inflating: mitdb/231.hea
  inflating: mitdb/118.atr
  inflating: mitdb/122.hea
  inflating: mitdb/223.dat
  inflating: mitdb/232.xws
  inflating: mitdb/212.atr
  inflating: mitdb/228.atr
  inflating: mitdb/228.xws
  inflating: mitdb/116.xws
  inflating: mitdb/207.dat
  inflating: mitdb/228.dat
  inflating: mitdb/201.hea
  inflating: mitdb/117.xws
  inflating: mitdb/228.hea
  inflating: mitdb/200.hea
  inflating: mitdb/213.hea
  inflating: mitdb/109.dat
  inflating: mitdb/222.hea
  inflating: mitdb/102.xws
  inflating: mitdb/205.xws
  inflating: mitdb/105.dat
  inflating: mitdb/100.atr
  inflating: mitdb/115.atr
  inflating: mitdb/230.xws
  inflating: mitdb/122.atr
  inflating: mitdb/222.xws
```

```
  inflating: mitdb/114.hea
  inflating: mitdb/100.hea
  inflating: mitdb/123.dat
  inflating: mitdb/210.hea
  inflating: mitdb/109.atr
  inflating: mitdb/SHA256SUMS.txt
  inflating: mitdb/220.atr
  inflating: mitdb/119.xws
  inflating: mitdb/203.dat
  inflating: mitdb/124.hea
  inflating: mitdb/103.hea
  inflating: mitdb/101.xws
  inflating: mitdb/234.atr
  inflating: mitdb/223.atr
  inflating: mitdb/209.hea
  inflating: mitdb/123.xws
  inflating: mitdb/214.at_
  inflating: mitdb/106.dat
  inflating: mitdb/114.xws
  inflating: mitdb/219.dat
  inflating: mitdb/230.hea
  inflating: mitdb/202.xws
  inflating: mitdb/105.atr
  inflating: mitdb/105.hea
   creating: mitdb/mitdbdir/
  inflating: mitdb/mitdbdir/index.html
   creating: mitdb/mitdbdir/src/
  inflating: mitdb/mitdbdir/src/notes.h
  inflating: mitdb/mitdbdir/src/domit
 extracting: mitdb/mitdbdir/src/notes.gz
  inflating: mitdb/mitdbdir/src/extext.tr
 extracting: mitdb/mitdbdir/src/fdtext.gz
  inflating: mitdb/mitdbdir/src/fdlist
  inflating: mitdb/mitdbdir/src/cover.tr
  inflating: mitdb/mitdbdir/src/tab.3
  inflating: mitdb/mitdbdir/src/fdtext.tr
  inflating: mitdb/mitdbdir/src/intro.gz
  inflating: mitdb/mitdbdir/src/tables.gz
 extracting: mitdb/mitdbdir/src/tab.5
  inflating: mitdb/mitdbdir/src/dbnotes.c
 extracting: mitdb/mitdbdir/src/extext.gz
  inflating: mitdb/mitdbdir/src/index.tr
 extracting: mitdb/mitdbdir/src/contents.gz
  inflating: mitdb/mitdbdir/src/makefile
  inflating: mitdb/mitdbdir/src/dbtab.c
  inflating: mitdb/mitdbdir/src/exlist
  inflating: mitdb/mitdbdir/src/contents.tr
  inflating: mitdb/mitdbdir/src/title.dvi
 extracting: mitdb/mitdbdir/src/cover.gz
```

```
 inflating: mitdb/mitdbdir/src/printdir.bat
 inflating: mitdb/mitdbdir/src/dbnotes-html.c
extracting: mitdb/mitdbdir/src/extest
 inflating: mitdb/mitdbdir/src/tab.4
 inflating: mitdb/mitdbdir/src/tab.2
 inflating: mitdb/mitdbdir/src/intro.tr
 inflating: mitdb/mitdbdir/src/makefile-html
 inflating: mitdb/mitdbdir/src/makelinks.pl
 inflating: mitdb/mitdbdir/src/tab.1
extracting: mitdb/mitdbdir/src/title.gz
extracting: mitdb/mitdbdir/src/index.gz
 inflating: mitdb/mitdbdir/records.htm
 inflating: mitdb/mitdbdir/tables.htm
  creating: mitdb/mitdbdir/samples/
extracting: mitdb/mitdbdir/samples/1052808.xws
extracting: mitdb/mitdbdir/samples/2322247.xws
extracting: mitdb/mitdbdir/samples/1032213.xws
extracting: mitdb/mitdbdir/samples/2342126.xws
extracting: mitdb/mitdbdir/samples/1052645.xws
extracting: mitdb/mitdbdir/samples/2002052.xws
extracting: mitdb/mitdbdir/samples/1140435.xws
extracting: mitdb/mitdbdir/samples/2022935.xws
extracting: mitdb/mitdbdir/samples/1001103.xws
extracting: mitdb/mitdbdir/samples/1030109.xws
extracting: mitdb/mitdbdir/samples/2021016.xws
extracting: mitdb/mitdbdir/samples/1020055.xws
extracting: mitdb/mitdbdir/samples/1050757.xws
extracting: mitdb/mitdbdir/samples/2070045.xws
extracting: mitdb/mitdbdir/samples/1091921.xws
extracting: mitdb/mitdbdir/samples/2202029.xws
extracting: mitdb/mitdbdir/samples/1092910.xws
extracting: mitdb/mitdbdir/samples/1130420.xws
extracting: mitdb/mitdbdir/samples/1052907.xws
extracting: mitdb/mitdbdir/samples/1110831.xws
extracting: mitdb/mitdbdir/samples/1162308.xws
extracting: mitdb/mitdbdir/samples/1040342.xws
extracting: mitdb/mitdbdir/samples/1232210.xws
extracting: mitdb/mitdbdir/samples/2052430.xws
extracting: mitdb/mitdbdir/samples/1182325.xws
extracting: mitdb/mitdbdir/samples/1020935.xws
extracting: mitdb/mitdbdir/samples/1060019.xws
extracting: mitdb/mitdbdir/samples/2192211.xws
extracting: mitdb/mitdbdir/samples/2051957.xws
extracting: mitdb/mitdbdir/samples/2000538.xws
extracting: mitdb/mitdbdir/samples/2330502.xws
extracting: mitdb/mitdbdir/samples/2170418.xws
extracting: mitdb/mitdbdir/samples/1210106.xws
extracting: mitdb/mitdbdir/samples/2140538.xws
extracting: mitdb/mitdbdir/samples/2282137.xws
```

```
extracting: mitdb/mitdbdir/samples/1232836.xws
extracting: mitdb/mitdbdir/samples/2232916.xws
extracting: mitdb/mitdbdir/samples/2150311.xws
extracting: mitdb/mitdbdir/samples/2192609.xws
extracting: mitdb/mitdbdir/samples/1010148.xws
 inflating: mitdb/mitdbdir/samples/2082808.xws
extracting: mitdb/mitdbdir/samples/2231907.xws
extracting: mitdb/mitdbdir/samples/2032651.xws
extracting: mitdb/mitdbdir/samples/1182858.xws
extracting: mitdb/mitdbdir/samples/2082858.xws
extracting: mitdb/mitdbdir/samples/1242741.xws
extracting: mitdb/mitdbdir/samples/2090047.xws
extracting: mitdb/mitdbdir/samples/2231755.xws
extracting: mitdb/mitdbdir/samples/1070044.xws
extracting: mitdb/mitdbdir/samples/2220645.xws
extracting: mitdb/mitdbdir/samples/1051516.xws
extracting: mitdb/mitdbdir/samples/2122438.xws
extracting: mitdb/mitdbdir/samples/1050527.xws
extracting: mitdb/mitdbdir/samples/2232547.xws
extracting: mitdb/mitdbdir/samples/2131730.xws
extracting: mitdb/mitdbdir/samples/2201041.xws
extracting: mitdb/mitdbdir/samples/2012620.xws
extracting: mitdb/mitdbdir/samples/1220109.xws
extracting: mitdb/mitdbdir/samples/2172227.xws
extracting: mitdb/mitdbdir/samples/2070128.xws
extracting: mitdb/mitdbdir/samples/2032715.xws
extracting: mitdb/mitdbdir/samples/2101757.xws
extracting: mitdb/mitdbdir/samples/2052418.xws
extracting: mitdb/mitdbdir/samples/1060137.xws
 inflating: mitdb/mitdbdir/samples/2070207.xws
extracting: mitdb/mitdbdir/samples/1091713.xws
extracting: mitdb/mitdbdir/samples/2330943.xws
extracting: mitdb/mitdbdir/samples/2202544.xws
extracting: mitdb/mitdbdir/samples/1082810.xws
extracting: mitdb/mitdbdir/samples/1062552.xws
extracting: mitdb/mitdbdir/samples/1080022.xws
extracting: mitdb/mitdbdir/samples/2082300.xws
extracting: mitdb/mitdbdir/samples/2081702.xws
extracting: mitdb/mitdbdir/samples/2092325.xws
extracting: mitdb/mitdbdir/samples/1012432.xws
extracting: mitdb/mitdbdir/samples/1002513.xws
extracting: mitdb/mitdbdir/samples/1170356.xws
extracting: mitdb/mitdbdir/samples/2010615.xws
extracting: mitdb/mitdbdir/samples/1091401.xws
extracting: mitdb/mitdbdir/samples/2222007.xws
extracting: mitdb/mitdbdir/samples/2330339.xws
extracting: mitdb/mitdbdir/samples/2302904.xws
extracting: mitdb/mitdbdir/samples/1031721.xws
extracting: mitdb/mitdbdir/samples/1090527.xws
```

```
extracting: mitdb/mitdbdir/samples/2140221.xws
extracting: mitdb/mitdbdir/samples/2070656.xws
extracting: mitdb/mitdbdir/samples/2022558.xws
extracting: mitdb/mitdbdir/samples/2231235.xws
extracting: mitdb/mitdbdir/samples/1150055.xws
extracting: mitdb/mitdbdir/samples/1152705.xws
extracting: mitdb/mitdbdir/samples/1150322.xws
extracting: mitdb/mitdbdir/samples/1082005.xws
extracting: mitdb/mitdbdir/samples/2012803.xws
extracting: mitdb/mitdbdir/samples/2341401.xws
extracting: mitdb/mitdbdir/samples/2192243.xws
extracting: mitdb/mitdbdir/samples/1142002.xws
extracting: mitdb/mitdbdir/samples/1002609.xws
extracting: mitdb/mitdbdir/samples/2202940.xws
extracting: mitdb/mitdbdir/samples/2330011.xws
extracting: mitdb/mitdbdir/samples/1240905.xws
extracting: mitdb/mitdbdir/samples/2152216.xws
extracting: mitdb/mitdbdir/samples/2092833.xws
extracting: mitdb/mitdbdir/samples/2001814.xws
extracting: mitdb/mitdbdir/samples/1212601.xws
extracting: mitdb/mitdbdir/samples/2012519.xws
extracting: mitdb/mitdbdir/samples/1061617.xws
extracting: mitdb/mitdbdir/samples/2140356.xws
extracting: mitdb/mitdbdir/samples/1061227.xws
extracting: mitdb/mitdbdir/samples/1130822.xws
extracting: mitdb/mitdbdir/samples/2211300.xws
extracting: mitdb/mitdbdir/samples/2300012.xws
extracting: mitdb/mitdbdir/samples/1052202.xws
extracting: mitdb/mitdbdir/samples/1010954.xws
extracting: mitdb/mitdbdir/samples/2122421.xws
extracting: mitdb/mitdbdir/samples/1180339.xws
extracting: mitdb/mitdbdir/samples/1160131.xws
extracting: mitdb/mitdbdir/samples/2002449.xws
extracting: mitdb/mitdbdir/samples/1121230.xws
extracting: mitdb/mitdbdir/samples/2321817.xws
extracting: mitdb/mitdbdir/samples/2321151.xws
extracting: mitdb/mitdbdir/samples/2070241.xws
extracting: mitdb/mitdbdir/samples/1232511.xws
extracting: mitdb/mitdbdir/samples/2092910.xws
extracting: mitdb/mitdbdir/samples/1090336.xws
extracting: mitdb/mitdbdir/samples/1020230.xws
extracting: mitdb/mitdbdir/samples/2340637.xws
extracting: mitdb/mitdbdir/samples/1010134.xws
extracting: mitdb/mitdbdir/samples/2232358.xws
extracting: mitdb/mitdbdir/samples/1082900.xws
extracting: mitdb/mitdbdir/samples/2021859.xws
extracting: mitdb/mitdbdir/samples/2070000.xws
extracting: mitdb/mitdbdir/samples/1031915.xws
extracting: mitdb/mitdbdir/samples/1092828.xws
```

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extracting: mitdb/mitdbdir/samples/1151552.xws
extracting: mitdb/mitdbdir/samples/2191957.xws
extracting: mitdb/mitdbdir/samples/2142317.xws
extracting: mitdb/mitdbdir/samples/1182648.xws
extracting: mitdb/mitdbdir/samples/1032333.xws
extracting: mitdb/mitdbdir/samples/2170651.xws
extracting: mitdb/mitdbdir/samples/2300511.xws
extracting: mitdb/mitdbdir/samples/2030500.xws
extracting: mitdb/mitdbdir/samples/1162547.xws
extracting: mitdb/mitdbdir/samples/2322353.xws
extracting: mitdb/mitdbdir/samples/1060423.xws
extracting: mitdb/mitdbdir/samples/2072910.xws
extracting: mitdb/mitdbdir/samples/1060757.xws
extracting: mitdb/mitdbdir/samples/1021612.xws
extracting: mitdb/mitdbdir/samples/2322334.xws
extracting: mitdb/mitdbdir/samples/2091421.xws
extracting: mitdb/mitdbdir/samples/2021845.xws
extracting: mitdb/mitdbdir/samples/1020451.xws
extracting: mitdb/mitdbdir/samples/2171530.xws
extracting: mitdb/mitdbdir/samples/1080741.xws
extracting: mitdb/mitdbdir/samples/2092817.xws
extracting: mitdb/mitdbdir/samples/2190508.xws
extracting: mitdb/mitdbdir/samples/1111153.xws
extracting: mitdb/mitdbdir/samples/2192855.xws
extracting: mitdb/mitdbdir/samples/1110404.xws
extracting: mitdb/mitdbdir/samples/2221948.xws
extracting: mitdb/mitdbdir/samples/2002831.xws
extracting: mitdb/mitdbdir/samples/1131148.xws
extracting: mitdb/mitdbdir/samples/2120005.xws
extracting: mitdb/mitdbdir/samples/2231351.xws
extracting: mitdb/mitdbdir/samples/2050457.xws
extracting: mitdb/mitdbdir/samples/1172227.xws
extracting: mitdb/mitdbdir/samples/2032325.xws
extracting: mitdb/mitdbdir/samples/2071735.xws
extracting: mitdb/mitdbdir/samples/2280019.xws
extracting: mitdb/mitdbdir/samples/2150255.xws
extracting: mitdb/mitdbdir/samples/2192443.xws
extracting: mitdb/mitdbdir/samples/1242322.xws
extracting: mitdb/mitdbdir/samples/2012233.xws
extracting: mitdb/mitdbdir/samples/2171656.xws
extracting: mitdb/mitdbdir/samples/2191350.xws
extracting: mitdb/mitdbdir/samples/2052757.xws
extracting: mitdb/mitdbdir/samples/1140926.xws
extracting: mitdb/mitdbdir/samples/2012016.xws
extracting: mitdb/mitdbdir/samples/1040617.xws
extracting: mitdb/mitdbdir/samples/2310002.xws
extracting: mitdb/mitdbdir/samples/1140356.xws
extracting: mitdb/mitdbdir/samples/1132210.xws
extracting: mitdb/mitdbdir/samples/2070519.xws
```

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extracting: mitdb/mitdbdir/samples/2342317.xws
extracting: mitdb/mitdbdir/samples/2322320.xws
extracting: mitdb/mitdbdir/samples/2002901.xws
extracting: mitdb/mitdbdir/samples/2010752.xws
extracting: mitdb/mitdbdir/samples/2090550.xws
extracting: mitdb/mitdbdir/samples/1182232.xws
extracting: mitdb/mitdbdir/samples/1090128.xws
extracting: mitdb/mitdbdir/samples/2152942.xws
extracting: mitdb/mitdbdir/samples/2171227.xws
extracting: mitdb/mitdbdir/samples/1190842.xws
extracting: mitdb/mitdbdir/samples/2022126.xws
extracting: mitdb/mitdbdir/samples/2211942.xws
extracting: mitdb/mitdbdir/samples/1240509.xws
extracting: mitdb/mitdbdir/samples/1222825.xws
extracting: mitdb/mitdbdir/samples/2321509.xws
extracting: mitdb/mitdbdir/samples/2341702.xws
extracting: mitdb/mitdbdir/samples/2010853.xws
 inflating: mitdb/mitdbdir/samples/1081808.xws
extracting: mitdb/mitdbdir/samples/2152457.xws
extracting: mitdb/mitdbdir/samples/2131538.xws
extracting: mitdb/mitdbdir/samples/2330218.xws
extracting: mitdb/mitdbdir/samples/1152128.xws
extracting: mitdb/mitdbdir/samples/1040513.xws
extracting: mitdb/mitdbdir/samples/2280050.xws
extracting: mitdb/mitdbdir/samples/2082106.xws
extracting: mitdb/mitdbdir/samples/1240521.xws
extracting: mitdb/mitdbdir/samples/2211356.xws
extracting: mitdb/mitdbdir/samples/2022110.xws
extracting: mitdb/mitdbdir/samples/2021241.xws
extracting: mitdb/mitdbdir/samples/2300926.xws
extracting: mitdb/mitdbdir/samples/2051522.xws
extracting: mitdb/mitdbdir/samples/2170033.xws
extracting: mitdb/mitdbdir/samples/2131448.xws
extracting: mitdb/mitdbdir/samples/2341426.xws
extracting: mitdb/mitdbdir/samples/1182953.xws
extracting: mitdb/mitdbdir/samples/2032202.xws
extracting: mitdb/mitdbdir/samples/2072714.xws
extracting: mitdb/mitdbdir/samples/2312110.xws
extracting: mitdb/mitdbdir/samples/2140235.xws
extracting: mitdb/mitdbdir/samples/2312011.xws
extracting: mitdb/mitdbdir/samples/1180831.xws
extracting: mitdb/mitdbdir/samples/2331620.xws
extracting: mitdb/mitdbdir/samples/1111541.xws
extracting: mitdb/mitdbdir/samples/1080813.xws
extracting: mitdb/mitdbdir/samples/2100656.xws
extracting: mitdb/mitdbdir/samples/2032404.xws
extracting: mitdb/mitdbdir/samples/1142929.xws
extracting: mitdb/mitdbdir/samples/1032858.xws
extracting: mitdb/mitdbdir/samples/2330243.xws
```

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extracting: mitdb/mitdbdir/samples/1241738.xws
extracting: mitdb/mitdbdir/samples/1152842.xws
extracting: mitdb/mitdbdir/samples/2212044.xws
extracting: mitdb/mitdbdir/samples/2140030.xws
extracting: mitdb/mitdbdir/samples/2051615.xws
extracting: mitdb/mitdbdir/samples/1062513.xws
extracting: mitdb/mitdbdir/samples/1002755.xws
extracting: mitdb/mitdbdir/samples/1182541.xws
extracting: mitdb/mitdbdir/samples/2200646.xws
extracting: mitdb/mitdbdir/samples/2051454.xws
extracting: mitdb/mitdbdir/samples/1240443.xws
extracting: mitdb/mitdbdir/samples/1232741.xws
extracting: mitdb/mitdbdir/samples/1051752.xws
extracting: mitdb/mitdbdir/samples/1141137.xws
extracting: mitdb/mitdbdir/samples/2282008.xws
extracting: mitdb/mitdbdir/samples/1020128.xws
extracting: mitdb/mitdbdir/samples/2222543.xws
extracting: mitdb/mitdbdir/samples/2211749.xws
extracting: mitdb/mitdbdir/samples/2051603.xws
extracting: mitdb/mitdbdir/samples/2280435.xws
extracting: mitdb/mitdbdir/samples/1110025.xws
extracting: mitdb/mitdbdir/samples/2231721.xws
extracting: mitdb/mitdbdir/samples/1140120.xws
extracting: mitdb/mitdbdir/samples/2222609.xws
extracting: mitdb/mitdbdir/samples/2171401.xws
extracting: mitdb/mitdbdir/samples/2281918.xws
extracting: mitdb/mitdbdir/samples/2132558.xws
extracting: mitdb/mitdbdir/samples/2002918.xws
 inflating: mitdb/mitdbdir/samples/1092609.xws
extracting: mitdb/mitdbdir/samples/2221732.xws
extracting: mitdb/mitdbdir/samples/1182623.xws
extracting: mitdb/mitdbdir/samples/2132856.xws
extracting: mitdb/mitdbdir/samples/2321306.xws
extracting: mitdb/mitdbdir/samples/2000142.xws
extracting: mitdb/mitdbdir/samples/2230935.xws
extracting: mitdb/mitdbdir/samples/2101555.xws
extracting: mitdb/mitdbdir/samples/2081457.xws
extracting: mitdb/mitdbdir/samples/2131755.xws
extracting: mitdb/mitdbdir/samples/2022213.xws
extracting: mitdb/mitdbdir/samples/1092803.xws
extracting: mitdb/mitdbdir/samples/1241027.xws
extracting: mitdb/mitdbdir/samples/2122846.xws
extracting: mitdb/mitdbdir/samples/2140508.xws
extracting: mitdb/mitdbdir/samples/2021822.xws
extracting: mitdb/mitdbdir/samples/2170000.xws
extracting: mitdb/mitdbdir/samples/1212432.xws
extracting: mitdb/mitdbdir/samples/2101314.xws
extracting: mitdb/mitdbdir/samples/2142752.xws
extracting: mitdb/mitdbdir/samples/1132901.xws
```

```
extracting: mitdb/mitdbdir/samples/2032639.xws
extracting: mitdb/mitdbdir/samples/2282642.xws
extracting: mitdb/mitdbdir/samples/1242603.xws
extracting: mitdb/mitdbdir/samples/2081910.xws
extracting: mitdb/mitdbdir/samples/1161232.xws
extracting: mitdb/mitdbdir/samples/2032446.xws
extracting: mitdb/mitdbdir/samples/2282149.xws
extracting: mitdb/mitdbdir/samples/2122645.xws
extracting: mitdb/mitdbdir/samples/1121140.xws
extracting: mitdb/mitdbdir/samples/2312322.xws
extracting: mitdb/mitdbdir/samples/2280038.xws
extracting: mitdb/mitdbdir/samples/1112702.xws
extracting: mitdb/mitdbdir/samples/1160044.xws
extracting: mitdb/mitdbdir/samples/2072536.xws
extracting: mitdb/mitdbdir/samples/1171158.xws
extracting: mitdb/mitdbdir/samples/2082937.xws
extracting: mitdb/mitdbdir/samples/2002612.xws
extracting: mitdb/mitdbdir/samples/2070425.xws
extracting: mitdb/mitdbdir/samples/2072610.xws
extracting: mitdb/mitdbdir/samples/1040822.xws
extracting: mitdb/mitdbdir/samples/1221347.xws
extracting: mitdb/mitdbdir/samples/2210000.xws
extracting: mitdb/mitdbdir/samples/2011814.xws
extracting: mitdb/mitdbdir/samples/2070057.xws
extracting: mitdb/mitdbdir/samples/2150946.xws
extracting: mitdb/mitdbdir/samples/1042651.xws
extracting: mitdb/mitdbdir/samples/2031502.xws
extracting: mitdb/mitdbdir/samples/2132651.xws
extracting: mitdb/mitdbdir/samples/2132443.xws
extracting: mitdb/mitdbdir/samples/2332223.xws
extracting: mitdb/mitdbdir/samples/1190238.xws
extracting: mitdb/mitdbdir/samples/1181347.xws
extracting: mitdb/mitdbdir/samples/2102915.xws
extracting: mitdb/mitdbdir/samples/1210333.xws
extracting: mitdb/mitdbdir/samples/2101732.xws
extracting: mitdb/mitdbdir/samples/2012350.xws
extracting: mitdb/mitdbdir/samples/2202358.xws
extracting: mitdb/mitdbdir/samples/1140831.xws
extracting: mitdb/mitdbdir/samples/1180923.xws
extracting: mitdb/mitdbdir/samples/1140536.xws
extracting: mitdb/mitdbdir/samples/2010000.xws
extracting: mitdb/mitdbdir/samples/1010513.xws
extracting: mitdb/mitdbdir/samples/2232951.xws
extracting: mitdb/mitdbdir/samples/2210243.xws
extracting: mitdb/mitdbdir/samples/1140000.xws
extracting: mitdb/mitdbdir/samples/1192005.xws
extracting: mitdb/mitdbdir/samples/1061052.xws
extracting: mitdb/mitdbdir/samples/1110241.xws
extracting: mitdb/mitdbdir/samples/2100347.xws
```

```
extracting: mitdb/mitdbdir/samples/1072038.xws
extracting: mitdb/mitdbdir/samples/2310224.xws
extracting: mitdb/mitdbdir/samples/2070033.xws
extracting: mitdb/mitdbdir/samples/2141353.xws
extracting: mitdb/mitdbdir/samples/2211912.xws
extracting: mitdb/mitdbdir/samples/1090013.xws
extracting: mitdb/mitdbdir/samples/1211648.xws
extracting: mitdb/mitdbdir/samples/2140325.xws
extracting: mitdb/mitdbdir/samples/2340016.xws
extracting: mitdb/mitdbdir/samples/2121842.xws
extracting: mitdb/mitdbdir/samples/1111954.xws
extracting: mitdb/mitdbdir/samples/1131227.xws
extracting: mitdb/mitdbdir/samples/1080451.xws
extracting: mitdb/mitdbdir/samples/2201810.xws
extracting: mitdb/mitdbdir/samples/1071230.xws
extracting: mitdb/mitdbdir/samples/2012415.xws
extracting: mitdb/mitdbdir/samples/2191900.xws
extracting: mitdb/mitdbdir/samples/1072552.xws
extracting: mitdb/mitdbdir/samples/1140339.xws
extracting: mitdb/mitdbdir/samples/1071954.xws
extracting: mitdb/mitdbdir/samples/1110352.xws
extracting: mitdb/mitdbdir/samples/2002951.xws
extracting: mitdb/mitdbdir/samples/1052727.xws
extracting: mitdb/mitdbdir/samples/1161637.xws
extracting: mitdb/mitdbdir/samples/1082420.xws
extracting: mitdb/mitdbdir/samples/2300000.xws
extracting: mitdb/mitdbdir/samples/2031314.xws
extracting: mitdb/mitdbdir/samples/2170705.xws
extracting: mitdb/mitdbdir/samples/2331802.xws
extracting: mitdb/mitdbdir/samples/1192533.xws
extracting: mitdb/mitdbdir/samples/2222443.xws
extracting: mitdb/mitdbdir/samples/2190249.xws
extracting: mitdb/mitdbdir/samples/2130339.xws
extracting: mitdb/mitdbdir/samples/1121249.xws
extracting: mitdb/mitdbdir/samples/1190451.xws
extracting: mitdb/mitdbdir/samples/1090446.xws
extracting: mitdb/mitdbdir/samples/2022755.xws
extracting: mitdb/mitdbdir/samples/2152757.xws
extracting: mitdb/mitdbdir/samples/2091257.xws
extracting: mitdb/mitdbdir/samples/2151558.xws
extracting: mitdb/mitdbdir/samples/1242013.xws
extracting: mitdb/mitdbdir/samples/1020112.xws
extracting: mitdb/mitdbdir/samples/1060253.xws
 inflating: mitdb/mitdbdir/samples/2222232.xws
extracting: mitdb/mitdbdir/samples/2320058.xws
extracting: mitdb/mitdbdir/samples/2170612.xws
extracting: mitdb/mitdbdir/samples/2131505.xws
extracting: mitdb/mitdbdir/samples/2152730.xws
extracting: mitdb/mitdbdir/samples/1042910.xws
extracting: mitdb/mitdbdir/samples/2170123.xws
```

```
extracting: mitdb/mitdbdir/samples/2152025.xws
extracting: mitdb/mitdbdir/samples/2330025.xws
extracting: mitdb/mitdbdir/samples/1040552.xws
extracting: mitdb/mitdbdir/samples/2021224.xws
extracting: mitdb/mitdbdir/samples/2102033.xws
extracting: mitdb/mitdbdir/samples/1081055.xws
extracting: mitdb/mitdbdir/samples/2190522.xws
extracting: mitdb/mitdbdir/samples/2320151.xws
extracting: mitdb/mitdbdir/samples/2231339.xws
extracting: mitdb/mitdbdir/samples/1161000.xws
extracting: mitdb/mitdbdir/samples/1190155.xws
extracting: mitdb/mitdbdir/samples/2322231.xws
 inflating: mitdb/mitdbdir/mitdbdir.htm
 inflating: mitdb/mitdbdir/foreword.htm
 inflating: mitdb/mitdbdir/intro.htm
 inflating: mitdb/104.xws
 inflating: mitdb/115.xws
 inflating: mitdb/209.xws
 inflating: mitdb/118.dat
 inflating: mitdb/108.hea
 inflating: mitdb/234.hea
 inflating: mitdb/200.dat
 inflating: mitdb/119.atr
 inflating: mitdb/122.xws
 inflating: mitdb/117.at_
 inflating: mitdb/100.xws
 inflating: mitdb/108.dat
 inflating: mitdb/123.hea
 inflating: mitdb/107.xws
 inflating: mitdb/119.dat
 inflating: mitdb/114.atr
 inflating: mitdb/210.atr
 inflating: mitdb/215.dat
 inflating: mitdb/122.dat
 inflating: mitdb/106.atr
 inflating: mitdb/221.hea
 inflating: mitdb/121.atr
 inflating: mitdb/104.dat
 inflating: mitdb/215.at_
 inflating: mitdb/201.atr
 inflating: mitdb/220.dat
 inflating: mitdb/100.dat
 inflating: mitdb/210.xws
 inflating: mitdb/209.dat
 inflating: mitdb/221.xws
 inflating: mitdb/223.hea
 inflating: mitdb/212.xws
 inflating: mitdb/111.atr
 inflating: mitdb/209.at_
```

```
  inflating: mitdb/104.hea
  inflating: mitdb/107.hea
  inflating: mitdb/234.xws
  inflating: mitdb/221.atr
  inflating: mitdb/115.hea
  inflating: mitdb/114.dat
  inflating: mitdb/119.at_
  inflating: mitdb/221.dat
  inflating: mitdb/124.dat
  inflating: mitdb/118.xws
   creating: mitdb/x_mitdb/
  inflating: mitdb/x_mitdb/x_221.dat
  inflating: mitdb/x_mitdb/x_232.hea
  inflating: mitdb/x_mitdb/x_222.hea
  inflating: mitdb/x_mitdb/x_109.hea
  inflating: mitdb/x_mitdb/x_124.hea
  inflating: mitdb/x_mitdb/x_114.hea
  inflating: mitdb/x_mitdb/x_112.dat
  inflating: mitdb/x_mitdb/RECORDS
  inflating: mitdb/x_mitdb/x_231.dat
  inflating: mitdb/x_mitdb/x_220.atr
  inflating: mitdb/x_mitdb/x_223.hea
  inflating: mitdb/x_mitdb/x_221.atr
  inflating: mitdb/x_mitdb/x_232.dat
  inflating: mitdb/x_mitdb/x_109.dat
  inflating: mitdb/x_mitdb/x_111.hea
 extracting: mitdb/x_mitdb/ANNOTATORS
  inflating: mitdb/x_mitdb/x_108.hea
  inflating: mitdb/x_mitdb/x_115.dat
  inflating: mitdb/x_mitdb/x_117.dat
  inflating: mitdb/x_mitdb/x_231.atr
  inflating: mitdb/x_mitdb/x_117.hea
  inflating: mitdb/x_mitdb/x_115.atr
  inflating: mitdb/x_mitdb/x_232.atr
  inflating: mitdb/x_mitdb/x_121.atr
  inflating: mitdb/x_mitdb/x_233.hea
  inflating: mitdb/x_mitdb/x_122.atr
  inflating: mitdb/x_mitdb/x_123.dat
  inflating: mitdb/x_mitdb/x_222.atr
  inflating: mitdb/x_mitdb/x_228.dat
  inflating: mitdb/x_mitdb/x_223.atr
  inflating: mitdb/x_mitdb/x_234.hea
  inflating: mitdb/x_mitdb/x_230.dat
  inflating: mitdb/x_mitdb/x_122.dat
  inflating: mitdb/x_mitdb/x_121.hea
  inflating: mitdb/x_mitdb/x_111.dat
  inflating: mitdb/x_mitdb/x_122.hea
  inflating: mitdb/x_mitdb/x_108.atr
  inflating: mitdb/x_mitdb/x_228.hea
```

```
  inflating: mitdb/x_mitdb/x_111.atr
  inflating: mitdb/x_mitdb/x_112.hea
  inflating: mitdb/x_mitdb/x_116.hea
  inflating: mitdb/x_mitdb/x_124.dat
  inflating: mitdb/x_mitdb/x_234.dat
  inflating: mitdb/x_mitdb/x_113.dat
  inflating: mitdb/x_mitdb/x_233.dat
  inflating: mitdb/x_mitdb/x_233.atr
  inflating: mitdb/x_mitdb/x_123.atr
  inflating: mitdb/x_mitdb/x_114.atr
  inflating: mitdb/x_mitdb/x_223.dat
  inflating: mitdb/x_mitdb/x_220.dat
  inflating: mitdb/x_mitdb/x_108.dat
  inflating: mitdb/x_mitdb/x_113.hea
  inflating: mitdb/x_mitdb/x_228.atr
  inflating: mitdb/x_mitdb/x_116.atr
  inflating: mitdb/x_mitdb/x_112.atr
  inflating: mitdb/x_mitdb/x_124.atr
  inflating: mitdb/x_mitdb/x_114.dat
  inflating: mitdb/x_mitdb/x_121.dat
  inflating: mitdb/x_mitdb/x_230.atr
  inflating: mitdb/x_mitdb/x_234.atr
  inflating: mitdb/x_mitdb/x_109.atr
  inflating: mitdb/x_mitdb/x_231.hea
  inflating: mitdb/x_mitdb/x_115.hea
  inflating: mitdb/x_mitdb/x_220.hea
  inflating: mitdb/x_mitdb/x_222.dat
  inflating: mitdb/x_mitdb/x_116.dat
  inflating: mitdb/x_mitdb/x_221.hea
  inflating: mitdb/x_mitdb/x_117.atr
  inflating: mitdb/x_mitdb/x_230.hea
  inflating: mitdb/x_mitdb/x_113.atr
  inflating: mitdb/x_mitdb/x_123.hea
  inflating: mitdb/101.dat
  inflating: mitdb/102.hea
  inflating: mitdb/233.dat
  inflating: mitdb/113.dat
  inflating: mitdb/103.atr
  inflating: mitdb/208.atr
  inflating: mitdb/205.hea
  inflating: mitdb/121.xws
  inflating: mitdb/116.hea
  inflating: mitdb/207.atr
  inflating: mitdb/213.xws
  inflating: mitdb/113.hea
  inflating: mitdb/217.hea
  inflating: mitdb/214.hea
  inflating: mitdb/106.hea
  inflating: mitdb/212.dat
```

```
  inflating: mitdb/102.atr
  inflating: mitdb/208.xws
  inflating: mitdb/200.xws
  inflating: mitdb/112.xws
  inflating: mitdb/124.xws
  inflating: mitdb/230.dat
  inflating: mitdb/113.xws
  inflating: mitdb/101.hea
  inflating: mitdb/214.atr
  inflating: mitdb/200.atr
  inflating: mitdb/231.dat
  inflating: mitdb/103.dat
  inflating: mitdb/108.xws
  inflating: mitdb/205.dat
  inflating: mitdb/214.xws
  inflating: mitdb/222.at_
  inflating: mitdb/202.dat
  inflating: mitdb/219.xws
  inflating: mitdb/123.atr
  inflating: mitdb/219.atr
  inflating: mitdb/201.dat
  inflating: mitdb/232.hea
  inflating: mitdb/124.atr
  inflating: mitdb/217.xws
  inflating: mitdb/232.dat
  inflating: mitdb/214.dat
```

```
!pip install wfdb tensorflow scikit-learn scipy pandas matplotlib
seaborn pywavelets

import os
import random
import warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

import wfdb
from scipy.signal import butter, filtfilt

from sklearn.model_selection import train_test_split
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    classification_report, confusion_matrix
)
from sklearn.utils.class_weight import compute_class_weight
```

```
import tensorflow as tf
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Input, SimpleRNN, LSTM, GRU,
Dense, Dropout
from tensorflow.keras.callbacks import EarlyStopping

SEED = 42
np.random.seed(SEED)
random.seed(SEED)
tf.random.set_seed(SEED)

print("TensorFlow:", tf.__version__)
print("WFDB:", wfdb.__version__)
```

```
Requirement already satisfied: wfdb in /usr/local/lib/python3.13/dist-
packages (4.3.1)
Requirement already satisfied: tensorflow in
/usr/local/lib/python3.13/dist-packages (2.21.0)
Requirement already satisfied: scikit-learn in
/usr/local/lib/python3.13/dist-packages (1.6.1)
Requirement already satisfied: scipy in
/usr/local/lib/python3.13/dist-packages (1.16.3)
Requirement already satisfied: pandas in
/usr/local/lib/python3.13/dist-packages (2.2.3)
Requirement already satisfied: matplotlib in
/usr/local/lib/python3.13/dist-packages (3.10.0)
Requirement already satisfied: seaborn in
/usr/local/lib/python3.13/dist-packages (0.13.2)
Requirement already satisfied: pywavelets in
/usr/local/lib/python3.13/dist-packages (1.10.0)
Requirement already satisfied: aiohttp>=3.10.11 in
/usr/local/lib/python3.13/dist-packages (from wfdb) (3.14.5)
Requirement already satisfied: fsspec>=2023.10.0 in
/usr/local/lib/python3.13/dist-packages (from wfdb) (2025.12.0)
Requirement already satisfied: numpy>=1.26.4 in
/usr/local/lib/python3.13/dist-packages (from wfdb) (2.1.3)
Requirement already satisfied: requests>=2.8.1 in
/usr/local/lib/python3.13/dist-packages (from wfdb) (2.32.4)
Requirement already satisfied: soundfile>=0.10.0 in
/usr/local/lib/python3.13/dist-packages (from wfdb) (0.14.0)
Requirement already satisfied: absl-py>=1.0.0 in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (2.0.0)
Requirement already satisfied: astunparse>=1.6.0 in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=25.9.23 in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1
in /usr/local/lib/python3.13/dist-packages (from tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (0.2.0)
```

```
Requirement already satisfied: libclang>=13.0.0 in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (18.1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (3.4.0)
Requirement already satisfied: packaging in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (26.3)
Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (6.33.6)
Requirement already satisfied: setuptools in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (80.10.2)
Requirement already satisfied: six>=1.12.0 in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (4.16.0)
Requirement already satisfied: wrapt>=1.11.0 in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (2.5.1)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (1.84.0)
Requirement already satisfied: keras>=3.12.0 in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (3.13.2)
Requirement already satisfied: h5py<3.15.0,>=3.11.0 in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (3.14.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in
/usr/local/lib/python3.13/dist-packages (from tensorflow) (0.6.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.13/dist-packages (from requests>=2.8.1->wfdb)
(3.4.9)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.13/dist-packages (from requests>=2.8.1->wfdb)
(3.20)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.13/dist-packages (from requests>=2.8.1->wfdb)
(2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.13/dist-packages (from requests>=2.8.1->wfdb)
(2026.10.19)
Requirement already satisfied: joblib>=1.2.0 in
/usr/local/lib/python3.13/dist-packages (from scikit-learn) (1.6.0)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.13/dist-packages (from scikit-learn) (3.7.0)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.13/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.13/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.13/dist-packages (from pandas) (2026.5)
Requirement already satisfied: contourpy>=1.0.1 in
```

```
/usr/local/lib/python3.13/dist-packages (from matplotlib) (1.4.0)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.13/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.13/dist-packages (from matplotlib) (4.67.0)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.13/dist-packages (from matplotlib) (1.5.1)
Requirement already satisfied: pillow>=8 in
/usr/local/lib/python3.13/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.13/dist-packages (from matplotlib) (3.3.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
/usr/local/lib/python3.13/dist-packages (from aiohttp>=3.10.11->wfdb)
(2.7.1)
Requirement already satisfied: aiosignal>=1.4.0 in
/usr/local/lib/python3.13/dist-packages (from aiohttp>=3.10.11->wfdb)
(1.4.0)
Requirement already satisfied: attrs>=17.3.0 in
/usr/local/lib/python3.13/dist-packages (from aiohttp>=3.10.11->wfdb)
(26.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in
/usr/local/lib/python3.13/dist-packages (from aiohttp>=3.10.11->wfdb)
(1.8.0)
Requirement already satisfied: multidict<8.0,>=4.5 in
/usr/local/lib/python3.13/dist-packages (from aiohttp>=3.10.11->wfdb)
(7.1.0)
Requirement already satisfied: propcache>=0.2.0 in
/usr/local/lib/python3.13/dist-packages (from aiohttp>=3.10.11->wfdb)
(0.5.4)
Requirement already satisfied: yarl<2.0,>=1.25.1 in
/usr/local/lib/python3.13/dist-packages (from aiohttp>=3.10.11->wfdb)
(1.25.1)
Requirement already satisfied: wheel<1.0,>=0.23.0 in
/usr/local/lib/python3.13/dist-packages (from astunparse>=1.6.0-
>tensorflow) (0.48.0)
Requirement already satisfied: cloudpickle>=3.0 in
/usr/local/lib/python3.13/dist-packages (from joblib>=1.2.0->scikit-
learn) (3.1.2)
Requirement already satisfied: rich in /usr/local/lib/python3.13/dist-
packages (from keras>=3.12.0->tensorflow) (13.9.4)
Requirement already satisfied: namex in
/usr/local/lib/python3.13/dist-packages (from keras>=3.12.0-
>tensorflow) (0.1.0)
Requirement already satisfied: optree in
/usr/local/lib/python3.13/dist-packages (from keras>=3.12.0-
>tensorflow) (0.20.0)
Requirement already satisfied: cffi>=1.0 in
/usr/local/lib/python3.13/dist-packages (from soundfile>=0.10.0->wfdb)
(2.1.1)
```

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Requirement already satisfied: pycparser in
/usr/local/lib/python3.13/dist-packages (from cffi>=1.0-
>soundfile>=0.10.0->wfdb) (3.11)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.13/dist-packages (from rich->keras>=3.12.0-
>tensorflow) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.13/dist-packages (from rich->keras>=3.12.0-
>tensorflow) (2.21.0)
Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.13/dist-packages (from markdown-it-py>=2.2.0-
>rich->keras>=3.12.0->tensorflow) (0.1.2)
TensorFlow: 2.21.0
WFDB: 4.3.1

DATA_DIR = "mitdb"
os.makedirs(DATA_DIR, exist_ok=True)

record_list = wfdb.get_record_list("mitdb")

for rec in record_list:
    try:
        wfdb.dl_database(
            "mitdb",
            dl_dir=DATA_DIR,
            records=[rec]
        )
    except Exception as e:
        print(f"Could not download {rec}: {e}")

print("Number of records:", len(record_list))
print(record_list[:10])

Generating record list for: 100
Generating list of all files for: 100
Downloading files...
Finished downloading files
Generating record list for: 101
Generating list of all files for: 101
Downloading files...
Finished downloading files
Generating record list for: 102
Generating list of all files for: 102
Downloading files...
Finished downloading files
Generating record list for: 103
Generating list of all files for: 103
Downloading files...
Finished downloading files
Generating record list for: 104
```

```
Generating list of all files for: 104
Downloading files...
Finished downloading files
Generating record list for: 105
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Finished downloading files
Number of records: 48
['100', '101', '102', '103', '104', '105', '106', '107', '108', '109']

record_name = record_list[0]
record_path = os.path.join(DATA_DIR, record_name)

record = wfdb.rdrecord(record_path)
annotation = wfdb.rdann(record_path, "atr")

print("Record:", record_name)
print("Sampling frequency:", record.fs)
print("Channels:", record.sig_name)
print("Signal shape:", record.p_signal.shape)

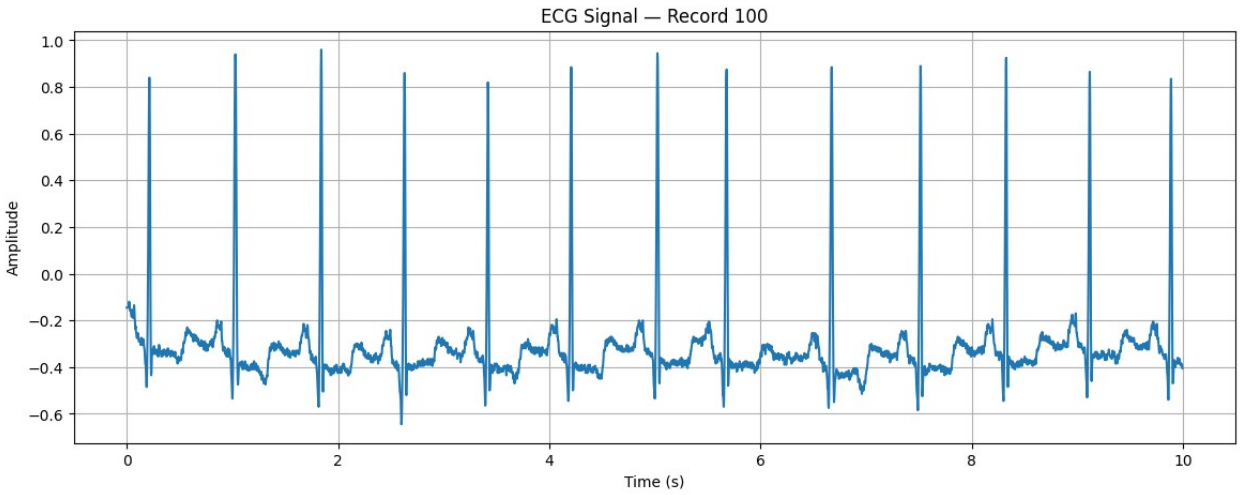
seconds = 10
n_samples = int(seconds * record.fs)

plt.figure(figsize=(14, 5))
plt.plot(np.arange(n_samples) / record.fs, record.p_signal[:n_samples,
0])
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
```

```
plt.title(f"ECG Signal — Record {record_name}")
plt.grid(True)
plt.show()

print("First annotation symbols:", annotation.symbol[:50])
```

```
Record: 100
Sampling frequency: 360
Channels: ['MLII', 'V5']
Signal shape: (650000, 2)
```




```
First annotation symbols: ['+', 'N', 'N', 'N', 'N', 'N', 'N', 'N',
'A', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N',
'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N',
'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N', 'N']
```

```
AAMI_MAP = {
    # N class
    "N": "N", "L": "N", "R": "N", "e": "N", "j": "N",

    # S class
    "A": "S", "a": "S", "J": "S", "S": "S",

    # V class
    "V": "V", "E": "V",

    # F class
    "F": "F",

    # Q class
    "/": "Q", "f": "Q", "Q": "Q"
}

# Beat symbols used for extracting labeled beats.
```

```
BEAT_SYMBOLS = set(AAMI_MAP.keys())

def record_has_paced_beats(symbols):
    return "/" in symbols

print("AAMI mapping entries:", len(AAMI_MAP))
```

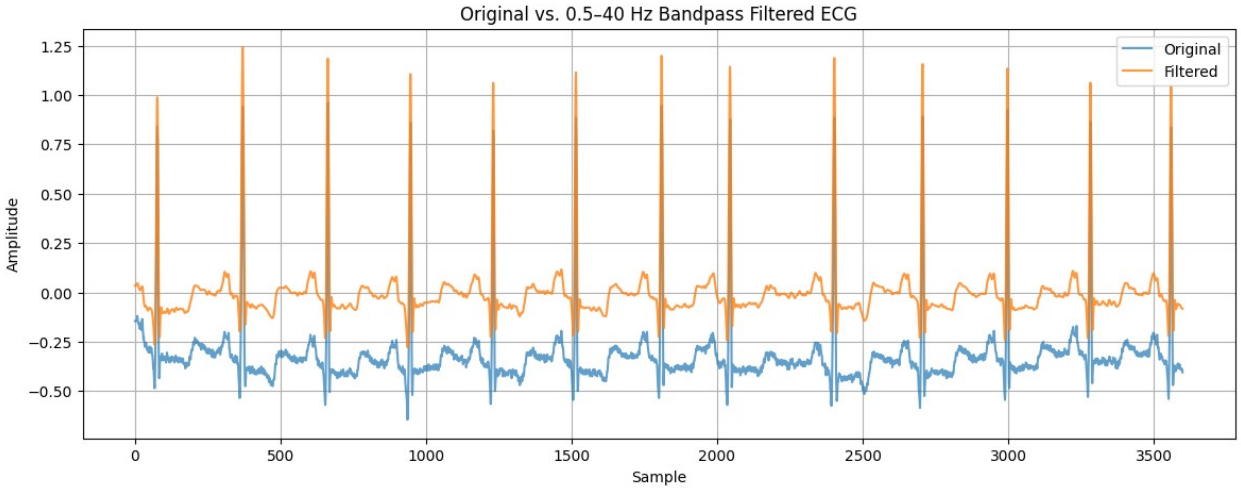
```
AAMI mapping entries: 15
```

```
FS = 360
LOWCUT = 0.5
HIGHCUT = 40.0

def bandpass_filter(signal, fs=FS, lowcut=LOWCUT, highcut=HIGHCUT,
order=4):
    nyquist = 0.5 * fs
    low = lowcut / nyquist
    high = highcut / nyquist
    b, a = butter(order, [low, high], btype="band")
    return filtfilt(b, a, signal)

filtered = bandpass_filter(record.p_signal[:, 0])

plt.figure(figsize=(14, 5))
plt.plot(record.p_signal[:n_samples, 0], label="Original", alpha=0.7)
plt.plot(filtered[:n_samples], label="Filtered", alpha=0.8)
plt.title("Original vs. 0.5–40 Hz Bandpass Filtered ECG")
plt.xlabel("Sample")
plt.ylabel("Amplitude")
plt.legend()
plt.grid(True)
plt.show()
```



```
BEFORE = 100
AFTER = 150
WINDOW_SIZE = BEFORE + AFTER

def extract_beats_from_record(rec_name, data_dir=DATA_DIR):
    path = os.path.join(data_dir, rec_name)

    rec = wfdb.rdrecord(path)
    ann = wfdb.rdann(path, "atr")

    # Use first ECG channel.
    signal = rec.p_signal[:, 0].astype(np.float32)
    signal = bandpass_filter(signal, fs=rec.fs)

    if record_has_paced_beats(ann.symbol):
        return None

    beats = []
    labels = []

    for sample, symbol in zip(ann.sample, ann.symbol):
        if symbol not in BEAT_SYMBOLS:
            continue

        label = AAMI_MAP[symbol]

        start = sample - BEFORE
        end = sample + AFTER

        if start < 0 or end > len(signal):
            continue

        beat = signal[start:end].copy()

        beat = (beat - np.mean(beat)) / (np.std(beat) + 1e-8)

        beats.append(beat)
        labels.append(label)

    if not beats:
        return None

    return np.asarray(beats, dtype=np.float32), np.asarray(labels),
rec_name

MAX_RECORDS = None

records_to_process = record_list if MAX_RECORDS is None else
record_list[:MAX_RECORDS]

all_X = []
```

```
all_y = []
all_groups = []
skipped = []

for i, rec_name in enumerate(records_to_process, 1):
    try:
        result = extract_beats_from_record(rec_name)

        if result is None:
            skipped.append(rec_name)
            continue

        X_rec, y_rec, group_rec = result

        all_X.append(X_rec)
        all_y.append(y_rec)
        all_groups.extend([group_rec] * len(y_rec))

        print(f"[{i}/{len(records_to_process)}] {rec_name}:
{len(y_rec)} beats")

    except Exception as e:
        skipped.append(rec_name)
        print(f"[{i}/{len(records_to_process)}] {rec_name}: ERROR ->
{e}")

X = np.concatenate(all_X, axis=0)
y = np.concatenate(all_y, axis=0)
groups = np.asarray(all_groups)

print("\nDataset shape:", X.shape)
print("Labels:", pd.Series(y).value_counts().sort_index())
print("Skipped records:", skipped)
```

```
[1/48] 100: 2271 beats
[2/48] 101: 1864 beats
[4/48] 103: 2083 beats
[6/48] 105: 2572 beats
[7/48] 106: 2027 beats
[9/48] 108: 1762 beats
[10/48] 109: 2531 beats
[11/48] 111: 2124 beats
[12/48] 112: 2538 beats
[13/48] 113: 1794 beats
[14/48] 114: 1879 beats
[15/48] 115: 1952 beats
[16/48] 116: 2411 beats
[17/48] 117: 1534 beats
[18/48] 118: 2277 beats
[19/48] 119: 1987 beats
```

```
[20/48] 121: 1862 beats
[21/48] 122: 2474 beats
[22/48] 123: 1517 beats
[23/48] 124: 1618 beats
[24/48] 200: 2600 beats
[25/48] 201: 1963 beats
[26/48] 202: 2135 beats
[27/48] 203: 2979 beats
[28/48] 205: 2655 beats
[29/48] 207: 1859 beats
[30/48] 208: 2953 beats
[31/48] 209: 3004 beats
[32/48] 210: 2648 beats
[33/48] 212: 2747 beats
[34/48] 213: 3249 beats
[35/48] 214: 2260 beats
[36/48] 215: 3362 beats
[38/48] 219: 2154 beats
[39/48] 220: 2046 beats
[40/48] 221: 2427 beats
[41/48] 222: 2481 beats
[42/48] 223: 2605 beats
[43/48] 228: 2053 beats
[44/48] 230: 2255 beats
[45/48] 231: 1570 beats
[46/48] 232: 1780 beats
[47/48] 233: 3077 beats
[48/48] 234: 2753 beats

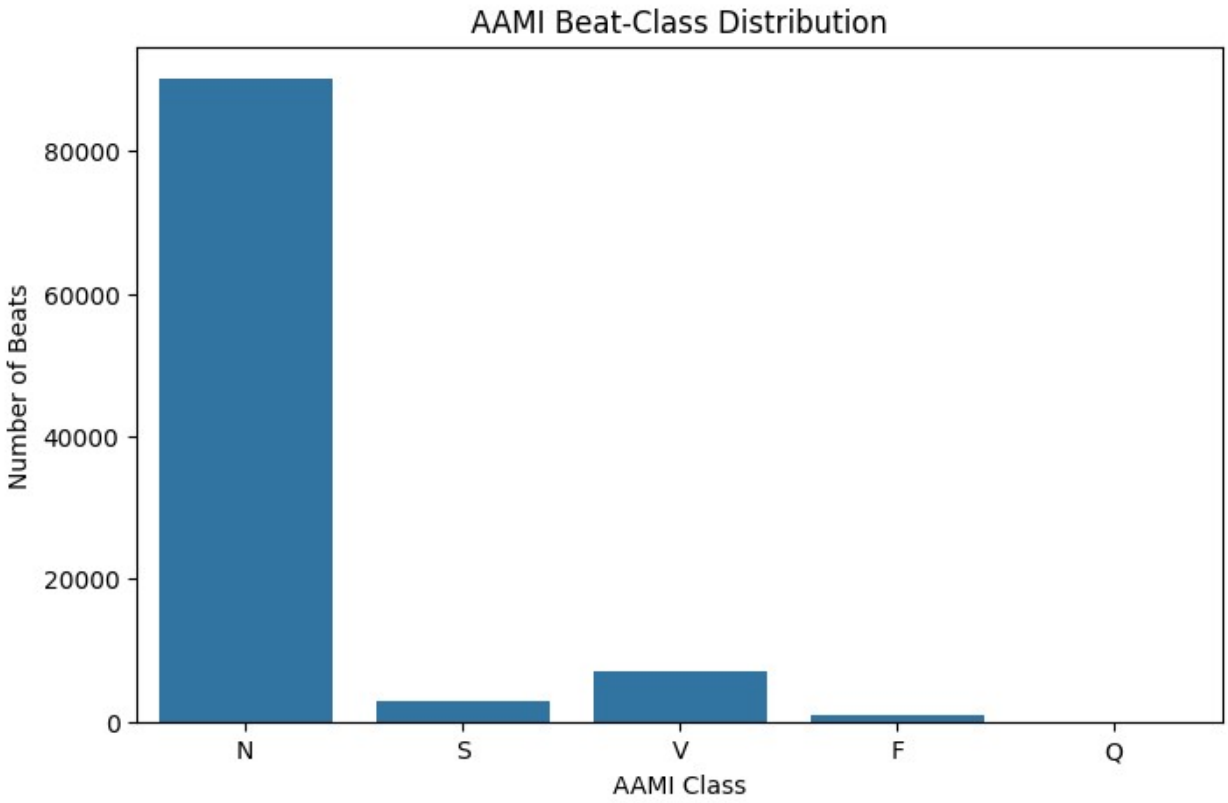
Dataset shape: (100692, 250)
Labels: F      802
N    90086
Q       15
S     2781
V     7008
Name: count, dtype: int64
Skipped records: ['102', '104', '107', '217']
```

```
class_counts = pd.Series(y).value_counts().reindex(["N", "S", "V",
"F", "Q"], fill_value=0)

display(class_counts.to_frame("Count"))

plt.figure(figsize=(8, 5))
sns.barplot(x=class_counts.index, y=class_counts.values)
plt.title("AAMI Beat-Class Distribution")
plt.xlabel("AAMI Class")
plt.ylabel("Number of Beats")
plt.show()
```

{"summary":"{\n \"name\": \"plt\",\n \"rows\": 5,\n \"fields\": [\n{\n   \"column\": \"Count\",\n   \"properties\": {\n\"dtype\": \"number\",\n   \"std\": 39195,\n   \"min\": 15,\n   \"max\": 90086,\n   \"num_unique_values\": 5,\n\"samples\": [\n   2781,\n   15,\n   7008\n],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n}\n   }\n]\n}","type":"dataframe"}




```
unique_groups = np.unique(groups)

train_groups, temp_groups = train_test_split(
    unique_groups,
    test_size=0.30,
    random_state=SEED
)

val_groups, test_groups = train_test_split(
    temp_groups,
    test_size=0.50,
    random_state=SEED
)

train_mask = np.isin(groups, train_groups)
val_mask = np.isin(groups, val_groups)
```

```
test_mask = np.isin(groups, test_groups)

X_train, y_train = X[train_mask], y[train_mask]
X_val, y_val = X[val_mask], y[val_mask]
X_test, y_test = X[test_mask], y[test_mask]

print("Train:", X_train.shape, len(y_train), "records:",
len(train_groups))
print("Validation:", X_val.shape, len(y_val), "records:",
len(val_groups))
print("Test:", X_test.shape, len(y_test), "records:",
len(test_groups))

print("\nTrain class distribution:")
print(pd.Series(y_train).value_counts().sort_index())

print("\nValidation class distribution:")
print(pd.Series(y_val).value_counts().sort_index())

print("\nTest class distribution:")
print(pd.Series(y_test).value_counts().sort_index())
```

```
Train: (68515, 250) 68515 records: 30
Validation: (15769, 250) 15769 records: 7
Test: (16408, 250) 16408 records: 7

Train class distribution:
F      769
N    60621
Q        8
S     1877
V     5240
Name: count, dtype: int64

Validation class distribution:
F       18
N    14628
S      338
V      785
Name: count, dtype: int64

Test class distribution:
F       15
N    14837
Q        7
S      566
V      983
Name: count, dtype: int64
```

```
CLASSES = ["N", "S", "V", "F", "Q"]
label_to_id = {label: i for i, label in enumerate(CLASSES)}
id_to_label = {i: label for label, i in label_to_id.items()}

y_train_int = np.array([label_to_id[v] for v in y_train])
y_val_int = np.array([label_to_id[v] for v in y_val])
y_test_int = np.array([label_to_id[v] for v in y_test])

present_classes = np.unique(y_train_int)
weights = compute_class_weight(
    class_weight="balanced",
    classes=present_classes,
    y=y_train_int
)
class_weight = {int(c): float(w) for c, w in zip(present_classes,
weights)}

print("Class weights:", class_weight)
```

```
Class weights: {0: 0.2260437802081787, 1: 7.300479488545552, 2:
2.6150763358778626, 3: 17.81924577373212, 4: 1712.875}
```

```
X_train_rnn = X_train[..., np.newaxis]
X_val_rnn = X_val[..., np.newaxis]
X_test_rnn = X_test[..., np.newaxis]

print("RNN input shape:", X_train_rnn.shape)
```

```
RNN input shape: (68515, 250, 1)
```

```
NUM_CLASSES = len(CLASSES)

def build_model(kind, input_shape, units=64):
    model = Sequential([
        Input(shape=input_shape)
    ])

    if kind == "RNN":
        model.add(SimpleRNN(units))
    elif kind == "LSTM":
        model.add(LSTM(units))
    elif kind == "GRU":
        model.add(GRU(units))
    else:
        raise ValueError("kind must be RNN, LSTM, or GRU")

    model.add(Dropout(0.3))
    model.add(Dense(NUM_CLASSES, activation="softmax"))

    model.compile(
        optimizer=tf.keras.optimizers.Adam(learning_rate=1e-3),
```

```
        loss="sparse_categorical_crossentropy",
        metrics=["accuracy"]
    )
    return model

models = {
    "RNN": build_model("RNN", X_train_rnn.shape[1:]),
    "LSTM": build_model("LSTM", X_train_rnn.shape[1:]),
    "GRU": build_model("GRU", X_train_rnn.shape[1:])
}

for name, model in models.items():
    print("\n", name)
    model.summary()
```

```
 RNN
Model: "sequential"
┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━┓
┃ Layer (type)                         ┃ Output Shape                ┃         Param # ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━┩
│ simple_rnn (SimpleRNN)               │ (None, 64)                  │           4,224 │
├──────────────────────────────────────┼─────────────────────────────┼─────────────────┤
│ dropout (Dropout)                    │ (None, 64)                  │               0 │
├──────────────────────────────────────┼─────────────────────────────┼─────────────────┤
│ dense (Dense)                        │ (None, 5)                   │             325 │
└──────────────────────────────────────┴─────────────────────────────┴─────────────────┘
 Total params: 4,549 (17.77 KB)
 Trainable params: 4,549 (17.77 KB)
 Non-trainable params: 0 (0.00 B)
 LSTM
Model: "sequential_1"
┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━┓
```

```
┃ Layer (type)                    ┃ Output Shape           ┃
Param # ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━
━━━━━┩
│ lstm (LSTM)                     │ (None, 64)             │
16,896 │
├─────────────────────────────────┼────────────────────────┼───────────
─────┤
│ dropout_1 (Dropout)             │ (None, 64)             │
0 │
├─────────────────────────────────┼────────────────────────┼───────────
─────┤
│ dense_1 (Dense)                 │ (None, 5)              │
325 │
└─────────────────────────────────┴────────────────────────┴───────────
─────┘

 Total params: 17,221 (67.27 KB)

 Trainable params: 17,221 (67.27 KB)

 Non-trainable params: 0 (0.00 B)


 GRU

Model: "sequential_2"

┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━
━━━━━┓
┃ Layer (type)                    ┃ Output Shape           ┃
Param # ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━
━━━━━┩
│ gru (GRU)                       │ (None, 64)             │
12,864 │
├─────────────────────────────────┼────────────────────────┼───────────
─────┤
│ dropout_2 (Dropout)             │ (None, 64)             │
0 │
├─────────────────────────────────┼────────────────────────┼───────────
─────┤
│ dense_2 (Dense)                 │ (None, 5)              │
325 │
└─────────────────────────────────┴────────────────────────┴───────────
─────┘

 Total params: 13,189 (51.52 KB)

 Trainable params: 13,189 (51.52 KB)
```

```
 Non-trainable params: 0 (0.00 B)
```

```
BATCH_SIZE = 128
EPOCHS = 30

histories = {}
trained_models = {}

for name, model in models.items():
    print(f"\n========== Training {name} ==========")

    early_stop = EarlyStopping(
        monitor="val_loss",
        patience=5,
        restore_best_weights=True,
        verbose=1
    )

    history = model.fit(
        X_train_rnn,
        y_train_int,
        validation_data=(X_val_rnn, y_val_int),
        epochs=EPOCHS,
        batch_size=BATCH_SIZE,
        class_weight=class_weight,
        callbacks=[early_stop],
        verbose=1
    )

    histories[name] = history
    trained_models[name] = model
```

```
========== Training RNN ==========
Epoch 1/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 21s 26ms/step - accuracy: 0.2405 - loss:
1.5878 - val_accuracy: 0.0351 - val_loss: 1.5726
Epoch 2/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 10s 18ms/step - accuracy: 0.2191 - loss:
1.6557 - val_accuracy: 0.3705 - val_loss: 1.4982
Epoch 3/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 10s 18ms/step - accuracy: 0.2139 - loss:
1.5823 - val_accuracy: 0.1487 - val_loss: 1.5742
Epoch 4/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 10s 18ms/step - accuracy: 0.2322 - loss:
1.4354 - val_accuracy: 0.1304 - val_loss: 1.5312
Epoch 5/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 10s 18ms/step - accuracy: 0.1837 - loss:
1.5900 - val_accuracy: 0.3391 - val_loss: 1.4390
Epoch 6/30
```

```
536/536 ━━━━━━━━━━━━━━━━━━━━ 10s 18ms/step - accuracy: 0.2036 - loss:
1.7552 - val_accuracy: 0.0651 - val_loss: 1.4169
Epoch 7/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 10s 18ms/step - accuracy: 0.2090 - loss:
1.6037 - val_accuracy: 0.6011 - val_loss: 1.2467
Epoch 8/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 10s 18ms/step - accuracy: 0.1897 - loss:
1.7520 - val_accuracy: 0.2646 - val_loss: 1.3213
Epoch 9/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 10s 19ms/step - accuracy: 0.1946 - loss:
1.7306 - val_accuracy: 0.2878 - val_loss: 1.4380
Epoch 10/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 10s 18ms/step - accuracy: 0.1838 - loss:
1.6158 - val_accuracy: 0.0474 - val_loss: 1.4881
Epoch 11/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 10s 18ms/step - accuracy: 0.2886 - loss:
1.5093 - val_accuracy: 0.3449 - val_loss: 1.1597
Epoch 12/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 10s 18ms/step - accuracy: 0.3159 - loss:
1.4635 - val_accuracy: 0.5862 - val_loss: 1.2624
Epoch 13/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 10s 19ms/step - accuracy: 0.2665 - loss:
1.5832 - val_accuracy: 0.9032 - val_loss: 1.3763
Epoch 14/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 10s 18ms/step - accuracy: 0.2334 - loss:
1.6711 - val_accuracy: 0.8033 - val_loss: 1.3523
Epoch 15/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 10s 18ms/step - accuracy: 0.2417 - loss:
1.6342 - val_accuracy: 0.7536 - val_loss: 1.3516
Epoch 16/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 10s 18ms/step - accuracy: 0.2325 - loss:
1.6191 - val_accuracy: 0.7312 - val_loss: 1.3470
Epoch 16: early stopping
Restoring model weights from the end of the best epoch: 11.

========== Training LSTM ==========
Epoch 1/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 12s 14ms/step - accuracy: 0.3160 - loss:
1.6795 - val_accuracy: 0.1152 - val_loss: 1.4127
Epoch 2/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 7s 13ms/step - accuracy: 0.1233 - loss:
1.4383 - val_accuracy: 0.2027 - val_loss: 1.4529
Epoch 3/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 7s 13ms/step - accuracy: 0.2164 - loss:
1.3192 - val_accuracy: 0.2170 - val_loss: 1.4771
Epoch 4/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 8s 14ms/step - accuracy: 0.2499 - loss:
1.3881 - val_accuracy: 0.2873 - val_loss: 1.4458
Epoch 5/30
```

```
536/536 ━━━━━━━━━━━━━━━━━━━━ 7s 13ms/step - accuracy: 0.2445 - loss:
1.5017 - val_accuracy: 0.3152 - val_loss: 1.4716
Epoch 6/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 7s 13ms/step - accuracy: 0.3979 - loss:
1.2614 - val_accuracy: 0.2859 - val_loss: 1.4614
Epoch 6: early stopping
Restoring model weights from the end of the best epoch: 1.

========== Training GRU ==========
Epoch 1/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 8s 13ms/step - accuracy: 0.4017 - loss:
1.6217 - val_accuracy: 0.3434 - val_loss: 1.4410
Epoch 2/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 7s 13ms/step - accuracy: 0.2160 - loss:
1.4749 - val_accuracy: 0.2251 - val_loss: 1.5525
Epoch 3/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 7s 13ms/step - accuracy: 0.1445 - loss:
1.5214 - val_accuracy: 0.1501 - val_loss: 1.5006
Epoch 4/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 7s 13ms/step - accuracy: 0.2768 - loss:
1.3014 - val_accuracy: 0.4436 - val_loss: 1.4432
Epoch 5/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 7s 12ms/step - accuracy: 0.3226 - loss:
1.2220 - val_accuracy: 0.4707 - val_loss: 1.3814
Epoch 6/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 7s 13ms/step - accuracy: 0.2905 - loss:
1.1568 - val_accuracy: 0.3567 - val_loss: 1.4121
Epoch 7/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 7s 12ms/step - accuracy: 0.2962 - loss:
1.2095 - val_accuracy: 0.3741 - val_loss: 1.3863
Epoch 8/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 7s 12ms/step - accuracy: 0.2938 - loss:
1.0584 - val_accuracy: 0.3110 - val_loss: 1.3945
Epoch 9/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 7s 13ms/step - accuracy: 0.2925 - loss:
1.1334 - val_accuracy: 0.2628 - val_loss: 1.4596
Epoch 10/30
536/536 ━━━━━━━━━━━━━━━━━━━━ 7s 12ms/step - accuracy: 0.2827 - loss:
1.0365 - val_accuracy: 0.2725 - val_loss: 1.4632
Epoch 10: early stopping
Restoring model weights from the end of the best epoch: 5.

fig, axes = plt.subplots(1, 2, figsize=(15, 5))

for name, history in histories.items():
    axes[0].plot(history.history["accuracy"], label=f"{name} train")
    axes[0].plot(history.history["val_accuracy"], "--", label=f"{name}
val")

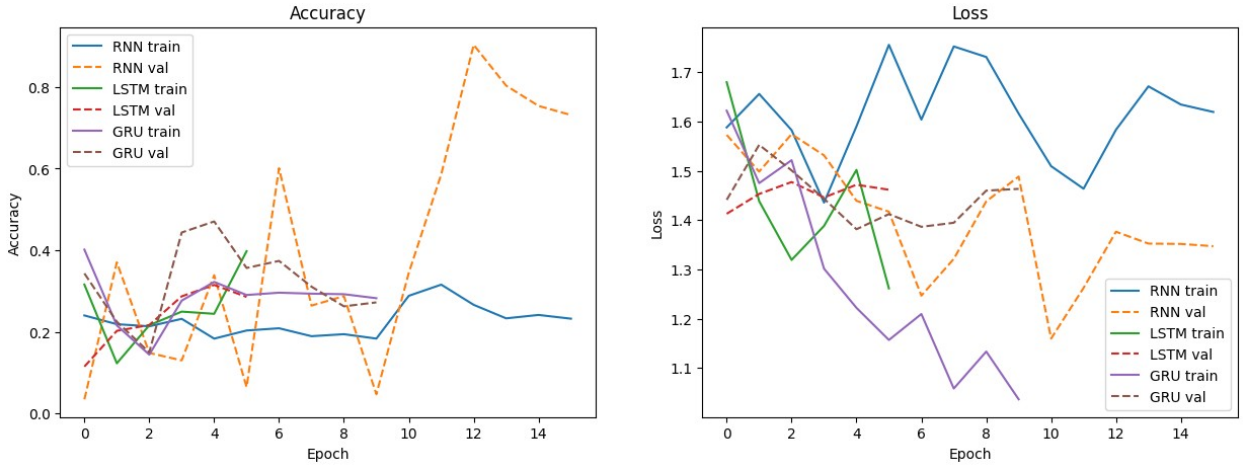
    axes[1].plot(history.history["loss"], label=f"{name} train")
```

```
    axes[1].plot(history.history["val_loss"], "--", label=f"{name}
val")

axes[0].set_title("Accuracy")
axes[0].set_xlabel("Epoch")
axes[0].set_ylabel("Accuracy")
axes[0].legend()

axes[1].set_title("Loss")
axes[1].set_xlabel("Epoch")
axes[1].set_ylabel("Loss")
axes[1].legend()

plt.show()
```




```
results = []
predictions = {}

for name, model in trained_models.items():
    prob = model.predict(X_test_rnn, batch_size=BATCH_SIZE, verbose=0)
    pred = np.argmax(prob, axis=1)
    predictions[name] = pred

    results.append({
        "Model": name,
        "Accuracy": accuracy_score(y_test_int, pred),
        "Precision (macro)": precision_score(y_test_int, pred,
average="macro", zero_division=0),
        "Recall (macro)": recall_score(y_test_int, pred,
average="macro", zero_division=0),
        "F1 (macro)": f1_score(y_test_int, pred, average="macro",
zero_division=0),
        "F1 (weighted)": f1_score(y_test_int, pred,
average="weighted", zero_division=0)
    })
```

```
results_df = pd.DataFrame(results).sort_values("F1 (macro)",
ascending=False)
display(results_df)
```


```
{"summary":"{\n  \"name\": \"results_df\",\n  \"rows\": 3,\n
\"fields\": [\n    {\n      \"column\": \"Model\",\n
\"properties\": {\n        \"dtype\": \"string\",\n
\"num_unique_values\": 3,\n        \"samples\": [\n
\"LSTM\",\n          \"GRU\",\n          \"RNN\"\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"Accuracy\",\n      \"properties\":
{\n        \"dtype\": \"number\",\n        \"std\":
0.058593063956566833,\n        \"min\": 0.10190151145782544,\n
\"max\": 0.2188566552901024,\n        \"num_unique_values\": 3,\n
\"samples\": [\n          0.2188566552901024,\n
0.15401023890784982,\n          0.10190151145782544\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"Precision (macro)\",\n
\"properties\": {\n        \"dtype\": \"number\",\n        \"std\":
0.011633324296834396,\n        \"min\": 0.20673431897309702,\n
\"max\": 0.22943068673367112,\n        \"num_unique_values\": 3,\n
\"samples\": [\n          0.22943068673367112,\n
0.21364867037532043,\n          0.20673431897309702\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"Recall (macro)\",\n
\"properties\": {\n        \"dtype\": \"number\",\n        \"std\":
0.066614049299262,\n        \"min\": 0.19636080291723085,\n
\"max\": 0.32319356859745463,\n        \"num_unique_values\": 3,\n
\"samples\": [\n          0.2244588922741829,\n
0.19636080291723085,\n          0.32319356859745463\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"F1 (macro)\",\n
\"properties\": {\n        \"dtype\": \"number\",\n        \"std\":
0.022691202616437808,\n        \"min\": 0.08733780990246404,\n
\"max\": 0.13154443509693747,\n        \"num_unique_values\": 3,\n
\"samples\": [\n          0.13154443509693747,\n
0.10055277365263068,\n          0.08733780990246404\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"F1 (weighted)\",\n
\"properties\": {\n        \"dtype\": \"number\",\n        \"std\":
0.09356124564083916,\n        \"min\": 0.13683006798144426,\n
\"max\": 0.3239475989675311,\n        \"num_unique_values\": 3,\n
\"samples\": [\n          0.3239475989675311,\n
0.23156877239216436,\n          0.13683006798144426\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    }\n  ]\n}","type":"dataframe","variable_name":"results_df"}
```


```
for name, pred in predictions.items():
    print("\n" + "=" * 70)
```

```
    print(name)
    print("=" * 70)
    print(
        classification_report(
            y_test_int,
            pred,
            labels=list(range(NUM_CLASSES)),
            target_names=CLASSES,
            zero_division=0
        )
    )
```

```
======================================================================
RNN
======================================================================
              precision    recall  f1-score   support

           N       0.83      0.07      0.13     14837
           S       0.00      0.00      0.00       566
           V       0.20      0.61      0.30       983
           F       0.00      0.93      0.00        15
           Q       0.00      0.00      0.00         7

    accuracy                           0.10     16408
   macro avg       0.21      0.32      0.09     16408
weighted avg       0.76      0.10      0.14     16408


======================================================================
LSTM
======================================================================
              precision    recall  f1-score   support

           N       0.86      0.21      0.34     14837
           S       0.07      0.48      0.13       566
           V       0.21      0.17      0.19       983
           F       0.00      0.27      0.00        15
           Q       0.00      0.00      0.00         7

    accuracy                           0.22     16408
   macro avg       0.23      0.22      0.13     16408
weighted avg       0.79      0.22      0.32     16408

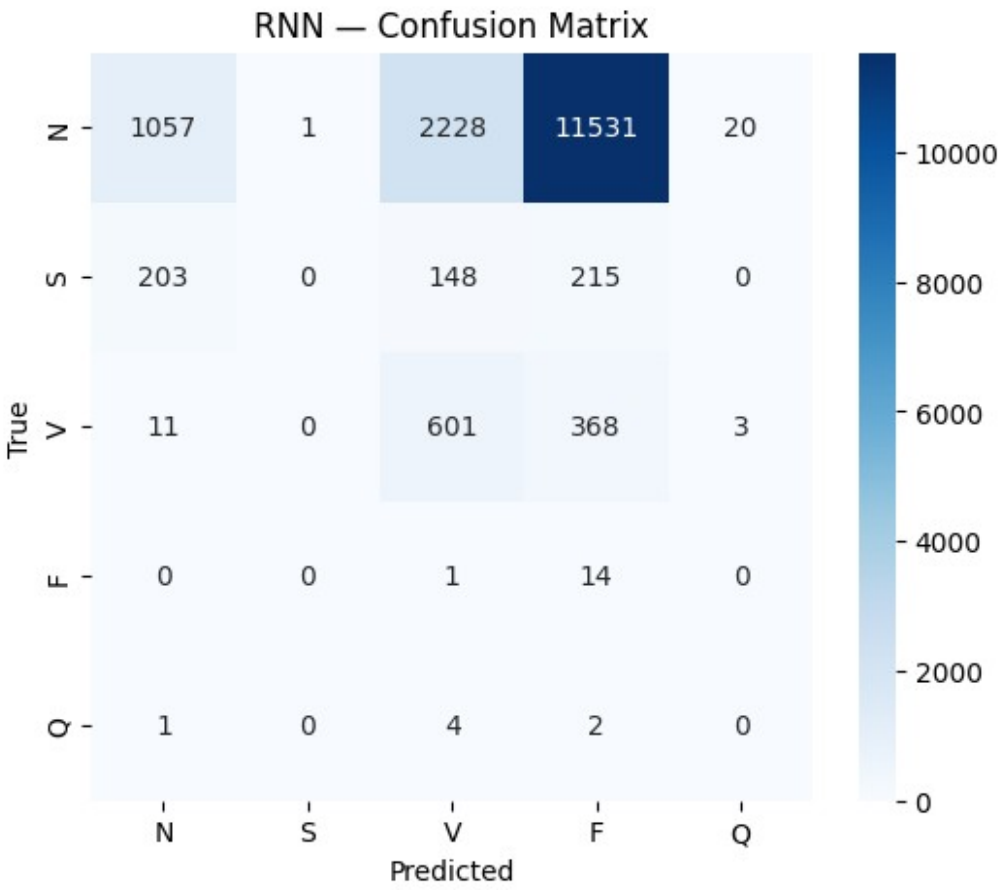

======================================================================
GRU
======================================================================
              precision    recall  f1-score   support
```

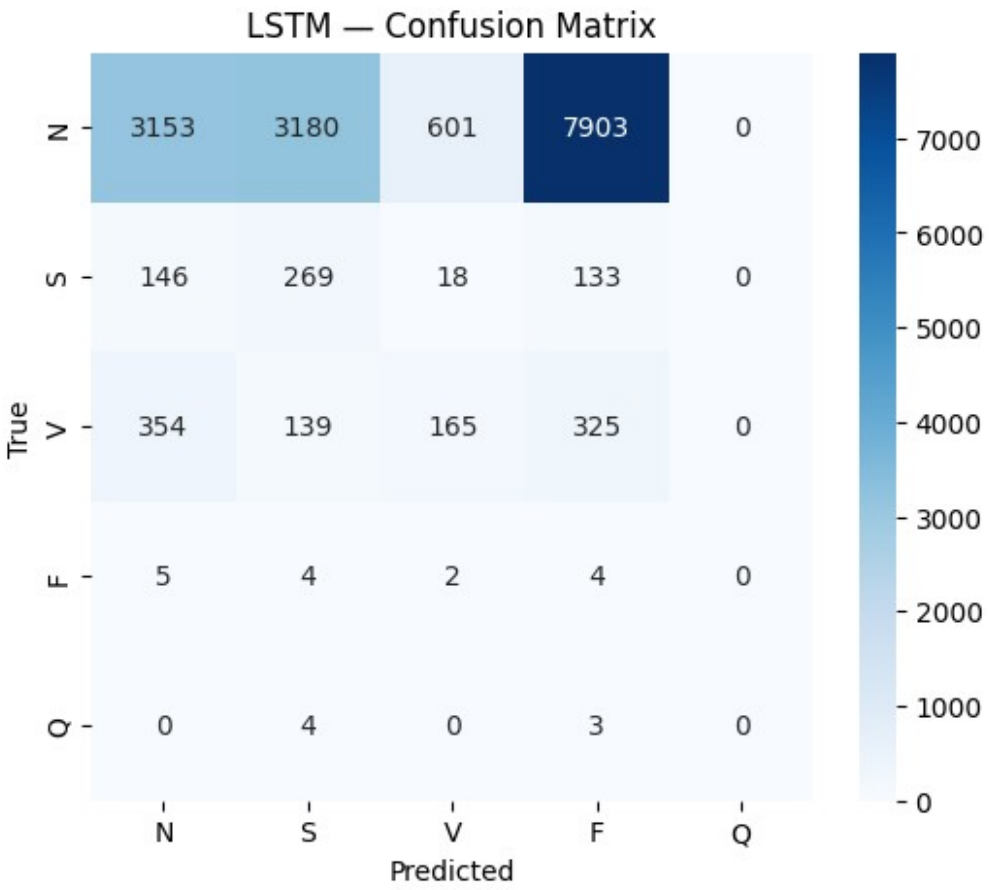
```
           N       0.84      0.14      0.24     14837
           S       0.18      0.17      0.17       566
           V       0.05      0.32      0.08       983
           F       0.00      0.07      0.00        15
           Q       0.00      0.29      0.00         7

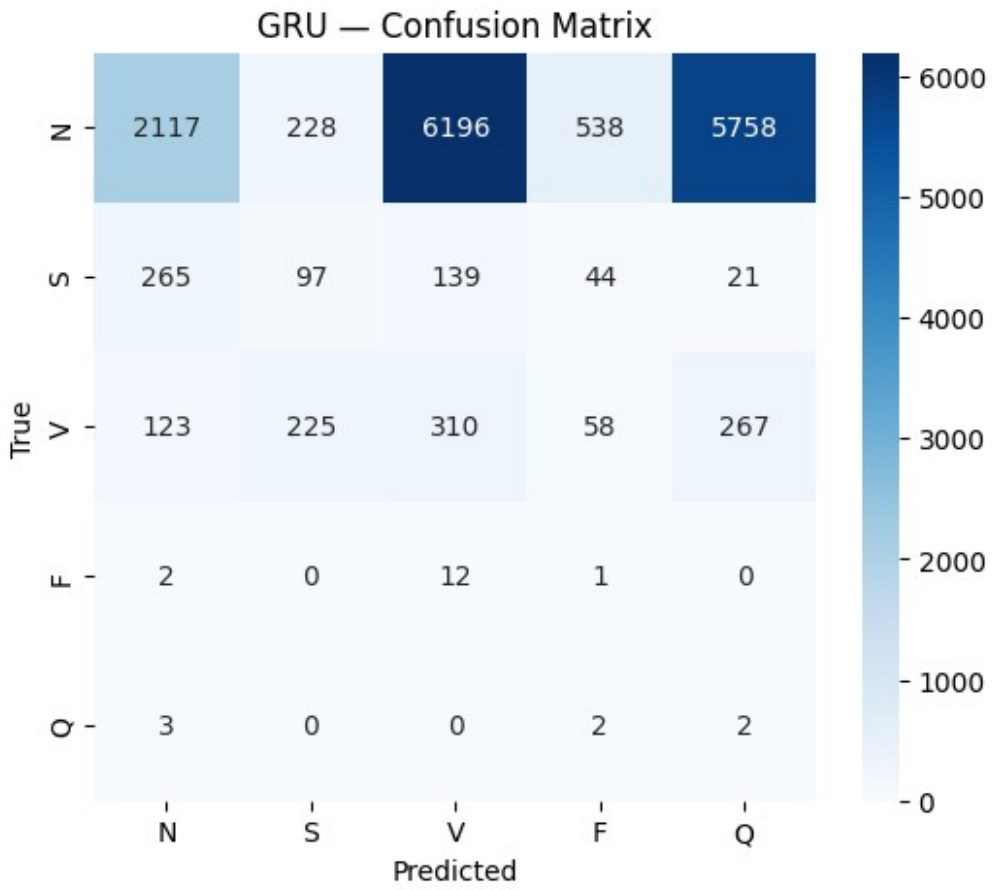
    accuracy                           0.15     16408
   macro avg       0.21      0.20      0.10     16408
weighted avg       0.77      0.15      0.23     16408
```

```python
for name, pred in predictions.items():
    cm = confusion_matrix(
        y_test_int,
        pred,
        labels=list(range(NUM_CLASSES))
    )

    plt.figure(figsize=(6, 5))
    sns.heatmap(
        cm,
        annot=True,
        fmt="d",
        xticklabels=CLASSES,
        yticklabels=CLASSES,
        cmap="Blues"
    )
    plt.xlabel("Predicted")
    plt.ylabel("True")
    plt.title(f"{name} — Confusion Matrix")
    plt.show()
```


RNN — Confusion Matrix
True
N
S
V
F
Q
1057
1
2228
11531
20
203
0
148
215
0
11
0
601
368
3
0
0
1
14
0
1
0
4
2
0
N
S
V
F
Q
Predicted
10000
8000
6000
4000
2000
0


LSTM — Confusion Matrix
True
N
S
V
F
Q
3153
3180
601
7903
0
146
269
18
133
0
354
139
165
325
0
5
4
2
4
0
0
4
0
3
0
N
S
V
F
Q
Predicted
7000
6000
5000
4000
3000
2000
1000
0




```
comparison = results_df.set_index("Model")
display(comparison)

best_model = results_df.iloc[0]["Model"]
print(f"Best model by macro F1: {best_model}")
```


```
{"summary":"{\n  \"name\": \"comparison\",\n  \"rows\": 3,\n
\"fields\": [\n    {\n      \"column\": \"Model\",\n
\"properties\": {\n        \"dtype\": \"string\",\n
\"num_unique_values\": 3,\n        \"samples\": [\n
\"LSTM\",\n          \"GRU\",\n          \"RNN\"\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"Accuracy\",\n      \"properties\":
{\n        \"dtype\": \"number\",\n        \"std\":
0.058593063956566833,\n        \"min\": 0.10190151145782544,\n
\"max\": 0.2188566552901024,\n        \"num_unique_values\": 3,\n
\"samples\": [\n          0.2188566552901024,\n
0.15401023890784982,\n          0.10190151145782544\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"Precision (macro)\",\n
\"properties\": {\n        \"dtype\": \"number\",\n        \"std\":
0.011633324296834396,\n        \"min\": 0.20673431897309702,\n
```

\"max\": 0.22943068673367112,\n   \"num_unique_values\": 3,\n
\"samples\": [\n   0.22943068673367112,\n
0.21364867037532043,\n   0.20673431897309702\n   ],\n
\"semantic_type\": \"\",\n   \"description\": \"\"\n   }\
n   },\n   {\n   \"column\": \"Recall (macro)\",\n
\"properties\": {\n   \"dtype\": \"number\",\n   \"std\":
0.066614049299262,\n   \"min\": 0.19636080291723085,\n
\"max\": 0.32319356859745463,\n   \"num_unique_values\": 3,\n
\"samples\": [\n   0.2244588922741829,\n
0.19636080291723085,\n   0.32319356859745463\n   ],\n
\"semantic_type\": \"\",\n   \"description\": \"\"\n   }\
n   },\n   {\n   \"column\": \"F1 (macro)\",\n
\"properties\": {\n   \"dtype\": \"number\",\n   \"std\":
0.022691202616437808,\n   \"min\": 0.08733780990246404,\n
\"max\": 0.13154443509693747,\n   \"num_unique_values\": 3,\n
\"samples\": [\n   0.13154443509693747,\n
0.10055277365263068,\n   0.08733780990246404\n   ],\n
\"semantic_type\": \"\",\n   \"description\": \"\"\n   }\
n   },\n   {\n   \"column\": \"F1 (weighted)\",\n
\"properties\": {\n   \"dtype\": \"number\",\n   \"std\":
0.09356124564083916,\n   \"min\": 0.13683006798144426,\n
\"max\": 0.3239475989675311,\n   \"num_unique_values\": 3,\n
\"samples\": [\n   0.3239475989675311,\n
0.23156877239216436,\n   0.13683006798144426\n   ],\n
\"semantic_type\": \"\",\n   \"description\": \"\"\n   }\
n   }\n]\n}","type":"dataframe","variable_name":"comparison"}

```
Best model by macro F1: LSTM
```

```
fig, axes = plt.subplots(len(CLASSES), 1, figsize=(12, 12))

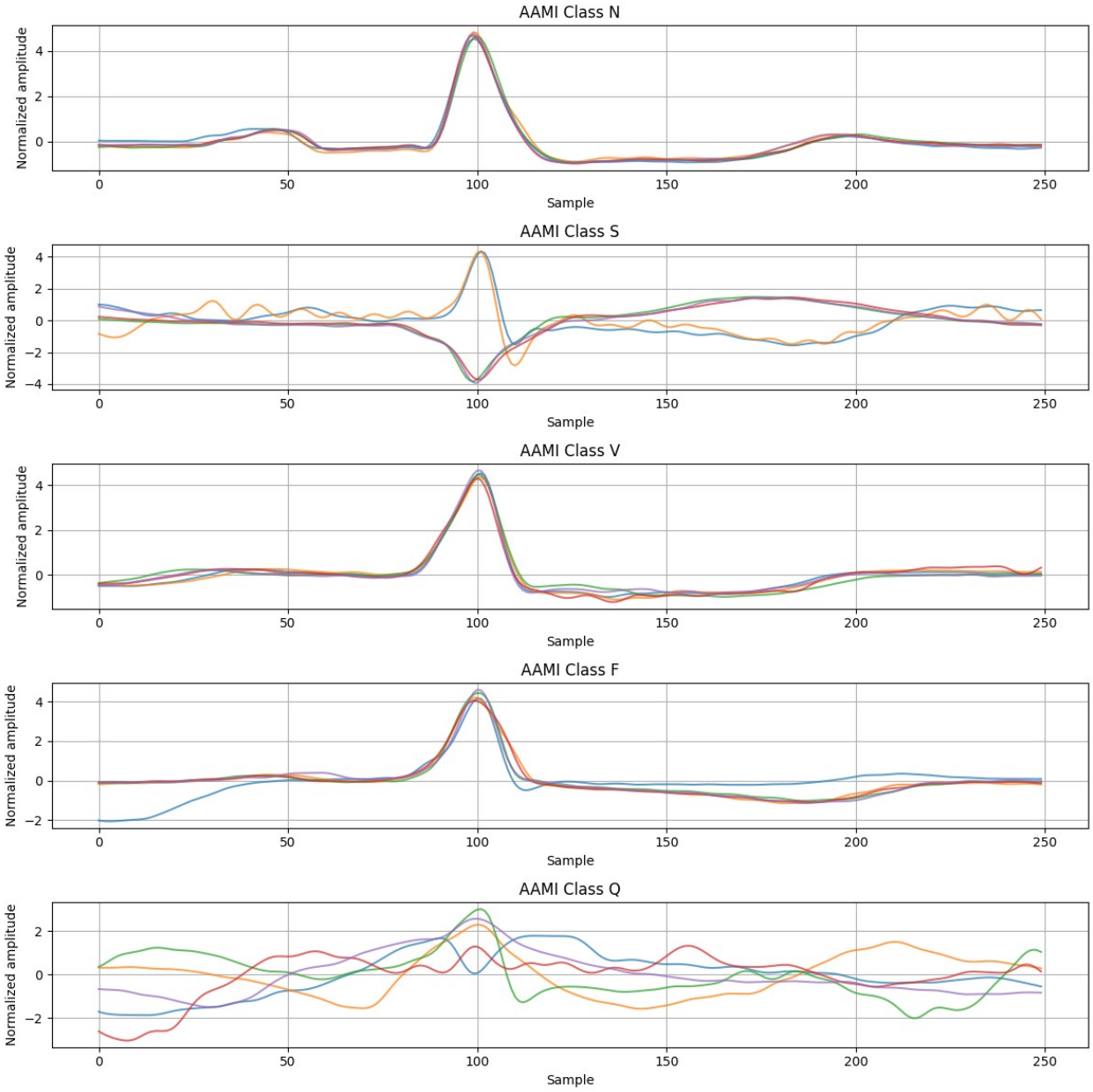
for ax, cls in zip(axes, CLASSES):
    idx = np.where(y_test == cls)[0]

    if len(idx) == 0:
        ax.set_title(f"No {cls} examples in test set")
        continue

    for j in idx[:5]:
        ax.plot(X_test[j], alpha=0.7)

    ax.set_title(f"AAMI Class {cls}")
    ax.set_xlabel("Sample")
    ax.set_ylabel("Normalized amplitude")
    ax.grid(True)

plt.tight_layout()
plt.show()
```


AAMI Class N
Normalized amplitude
Sample
AAMI Class S
Normalized amplitude
Sample
AAMI Class V
Normalized amplitude
Sample
AAMI Class F
Normalized amplitude
Sample
AAMI Class Q
Normalized amplitude
Sample